

1. Which of the following statements about *boiling* and *evaporation* of a liquid is/are correct ?

- (1) A liquid absorbs energy when it boils but does not absorb energy when it evaporates.
- (2) Boiling occurs at a definite temperature while evaporation takes place above room temperature.
- (3) Boiling occurs throughout the liquid while evaporation only takes place at the liquid's surface.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

2. In an experiment to measure the specific latent heat of vaporization of water, a beaker of water is boiled off using an electric heater. Which of the following sources of error would lead to an experimental result smaller than the standard value ?

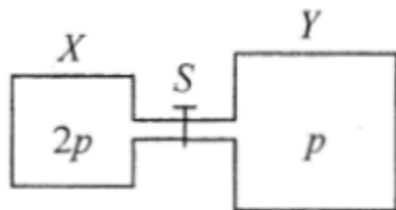
- A. Energy is lost to the surroundings.
- B. Water splashes out of the beaker.
- C. Steam condenses on the cooler part of the heater and drops back to the beaker.
- D. The heater is not completely immersed in water.

*3. In which of the following situations would the r.m.s. speed of the molecules of a fixed mass of an ideal gas increase ?

- (1) The gas is heated under constant volume.
- (2) The gas expands under constant pressure.
- (3) The gas is compressed under constant temperature.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

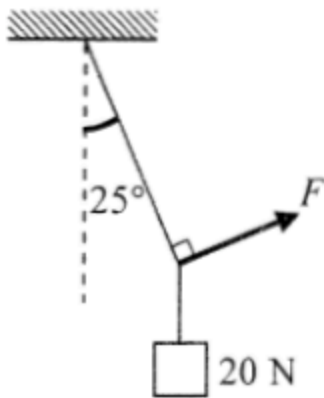
*4.



Vessel X of volume V and vessel Y of volume $2V$ are connected by a short narrow tube as shown. Initially, tap S is closed and the same kind of ideal gas at the same temperature is contained in X and Y at pressure $2p$ and p respectively. The tap S is then opened and equilibrium state is finally reached with the temperature unchanged. Which statement is **INCORRECT** ?

- A. Before S is opened, both vessels contain the same number of gas molecules.
- B. Before S is opened, the average kinetic energy of the gas molecules in both vessels is the same.
- C. When S is opened, a net flow of gas from X to Y occurs.
- D. When equilibrium is reached, the gas pressure becomes $\frac{3}{2}p$.

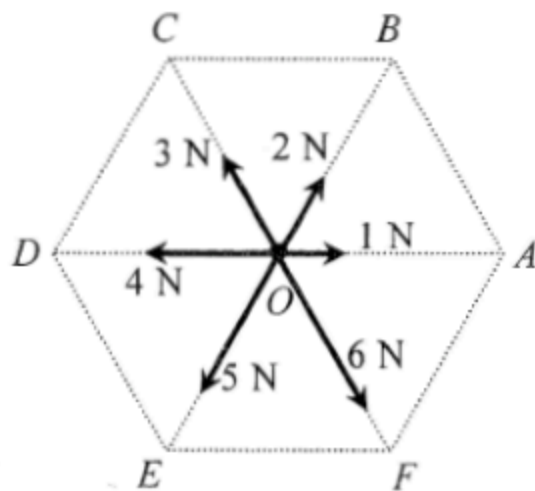
5.



A block of weight 20 N is suspended by a light string from the ceiling. A force F is applied such that the block is displaced to one side with the string making an angle of 25° with the vertical as shown. Find the magnitude of F .

- A. 8.5 N
- B. 9.3 N
- C. 18.1 N
- D. 47.3 N

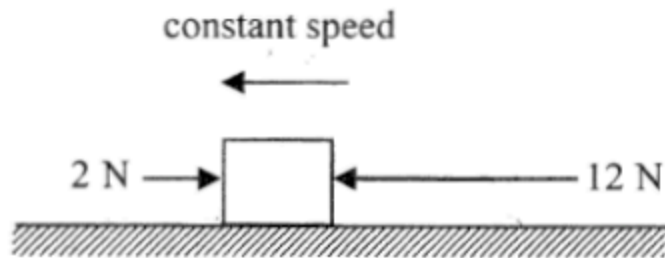
6.



In the figure, O is the centre of a regular hexagon. A particle at O is subject to six forces with magnitudes indicated as shown. The resultant force acting on the particle is

- A. 9 N along direction OE .
- B. 8 N along direction OE .
- C. 8 N along direction OF .
- D. 6 N along direction OE .

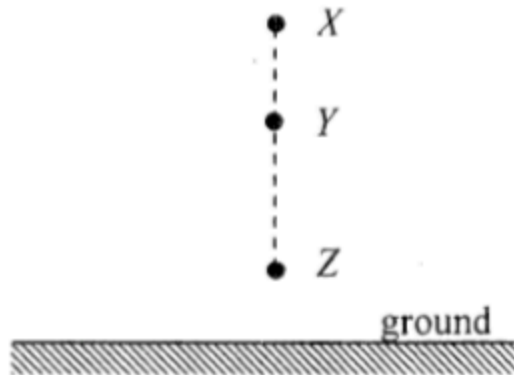
7.



A block on a rough horizontal surface is moving to the left with constant speed under two horizontal forces 2 N and 12 N indicated as shown. If the force of 12 N is suddenly removed, what is the net force acting on the block at that instant ?

- A. 12 N
- B. 10 N
- C. 8 N
- D. 2 N

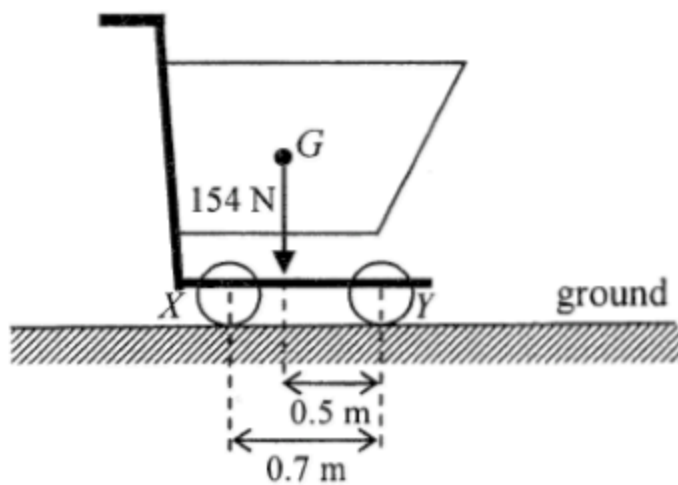
8.



A particle is released from rest at X as shown. It takes time t_1 to fall from X to Y and time t_2 to fall from Y to Z . If $XY : YZ = 9 : 16$, find $t_1 : t_2$. Neglect air resistance.

- A. 2 : 3
- B. 3 : 4
- C. 4 : 3
- D. 3 : 2

9.



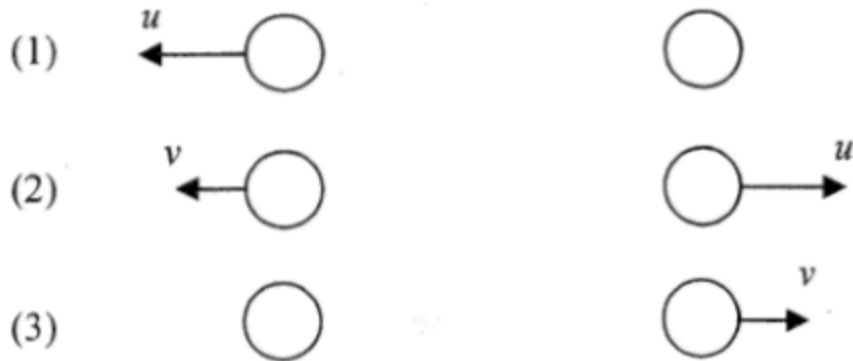
The figure shows a supermarket trolley resting on the ground. The separation between cylindrical wheels X and Y is 0.7 m . When the trolley is loaded to a total weight of 154 N , its centre of gravity G is at a horizontal distance of 0.5 m from the wheel Y . What is the reaction acting on the wheel X from the ground ?

- A. 44 N
- B. 62 N
- C. 92 N
- D. 110 N

10.



Two identical spheres are moving in opposite directions with speeds u and v (with $u > v$) respectively as shown. They make a head-on collision. Which of the following diagrams show(s) a *possible* situation of the spheres after collision?

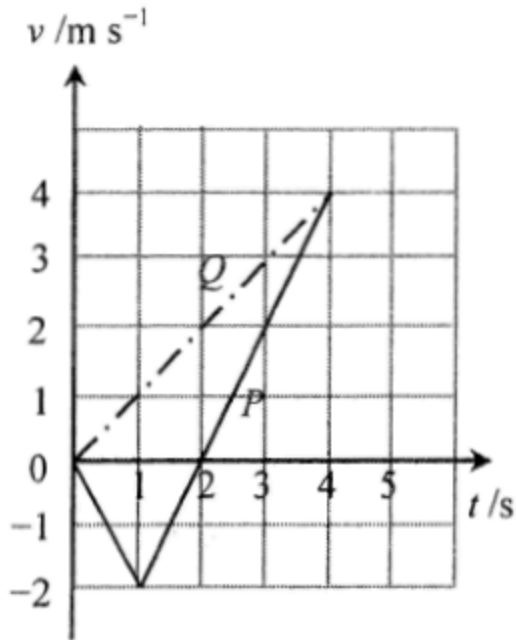


- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

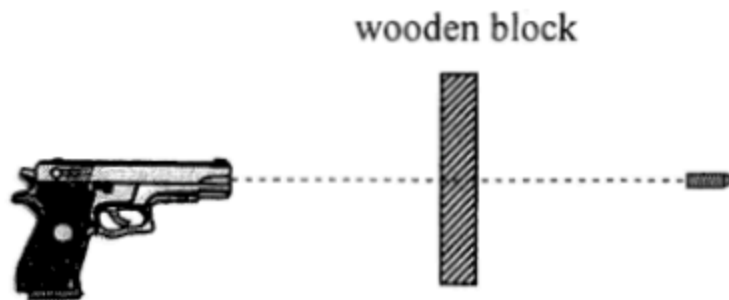
11. Two particles P and Q start from the same position and travel along the same straight line. The figure shows the velocity-time (v - t) graph for P and Q . Which of the following descriptions about their motion is/are correct?

- (1) At $t = 1$ s, P changes its direction of motion.
- (2) At $t = 2$ s, the separation between P and Q is 4 m.
- (3) At $t = 4$ s, P and Q meet each other.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only



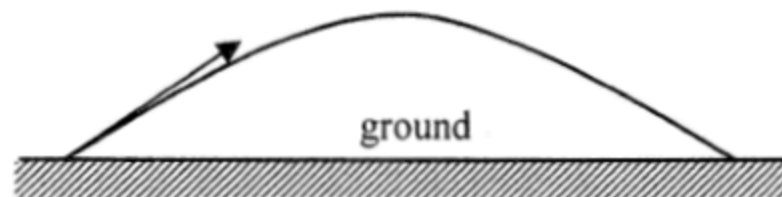
12.



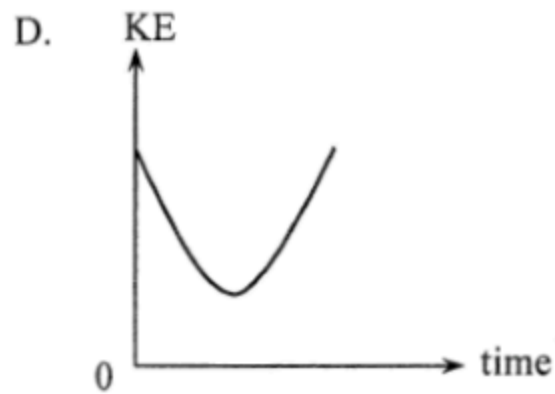
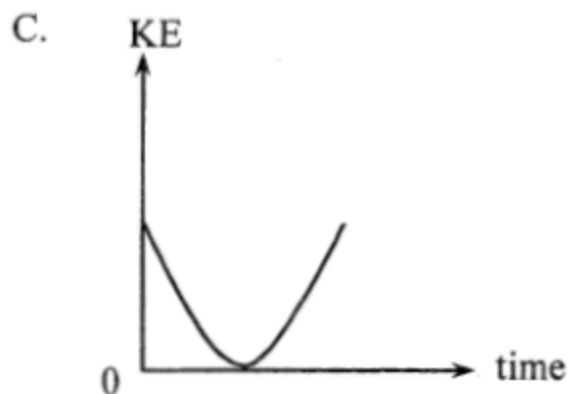
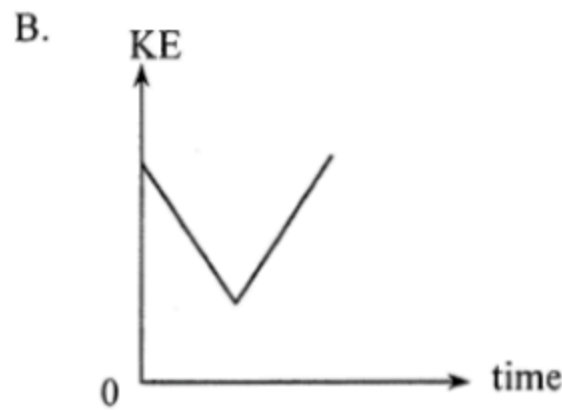
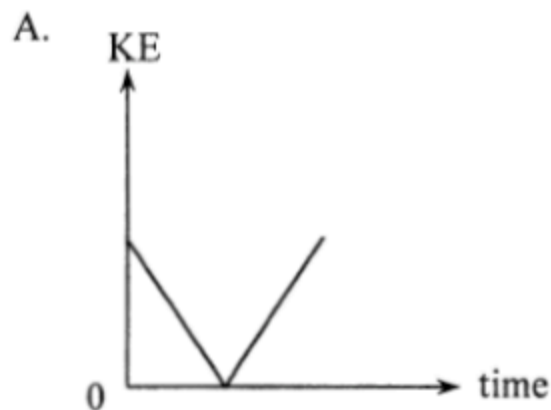
A bullet of mass 50 g is fired from a gun with a speed of 400 m s^{-1} and passes right through a fixed wooden block of 6 cm thickness as shown. Find the average resistive force acting on the bullet due to the block if it emerges with a speed of 250 m s^{-1} . Neglect air resistance and the effects of gravity.

- A. $4.06 \times 10^4 \text{ N}$
- B. $1.02 \times 10^4 \text{ N}$
- C. 125 N
- D. Answer cannot be found as the time of travel of the bullet within the block is not known.

*13.



A particle is projected into the air at time $t = 0$ and it performs a parabolic motion before landing on the ground as shown. Which graph represents the variation of the kinetic energy (KE) of the particle with time before landing? Neglect air resistance.





A semi-circular cardboard hangs from a spring balance from point O as shown. The reading of the spring balance is 5 N. Which statements are correct ?

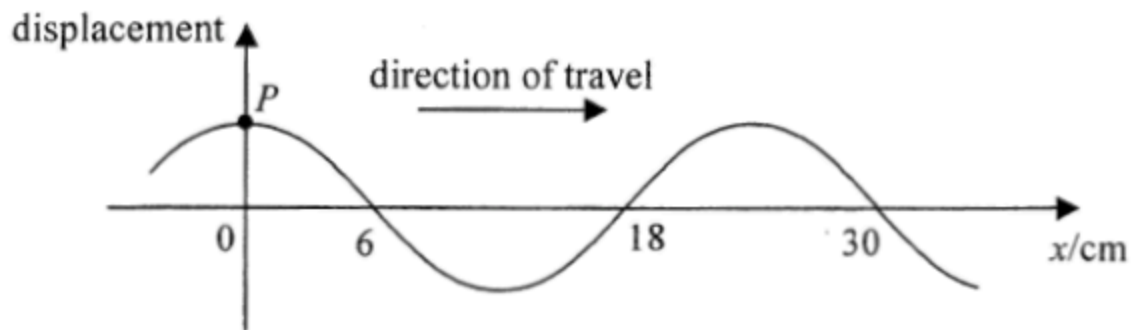
- (1) The weight of the cardboard is 5 N.
- (2) The centre of gravity of the cardboard is directly under O .
- (3) The reading of the balance would become zero if the set-up is brought to the Moon's surface.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- *15. It is known that the mass of Mars is about $\frac{1}{10}$ of that of the Earth while its radius is about $\frac{1}{2}$ of the Earth's radius. In terms of the gravitational acceleration g on the Earth's surface, the approximate gravitational acceleration on the surface of Mars is

- A. $0.2 g$.
- B. $0.4 g$.
- C. $2.5 g$.
- D. $4 g$.

16.



The figure shows a snapshot of a section of a continuous transverse wave travelling along the x -direction at time $t = 0$. At $t = 1.5$ s, the particle P just passes the equilibrium position for a *second* time at that moment. Find the wave speed.

- A. 20 cm s^{-1}
- B. 12 cm s^{-1}
- C. 6 cm s^{-1}
- D. 4 cm s^{-1}

17.

Figure (1)

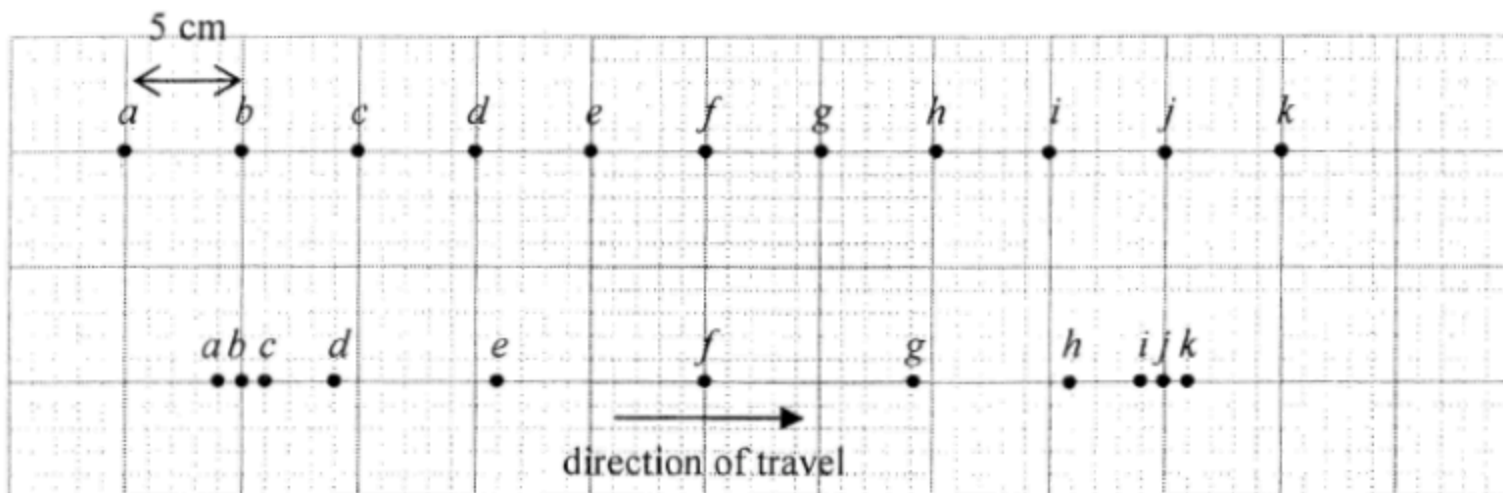
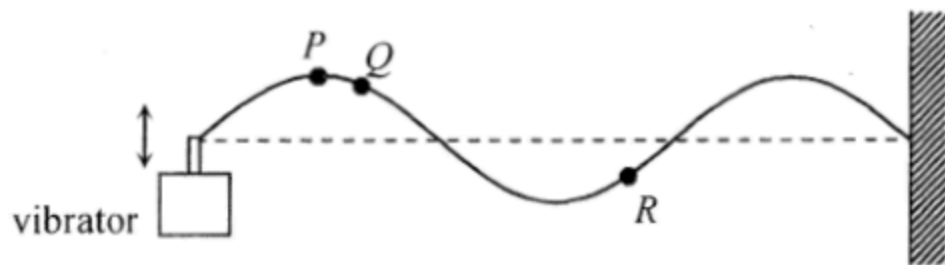


Figure (2)

Figure (1) shows the equilibrium positions of particles *a* to *k* separated by 5 cm from each other in a medium. A longitudinal wave is travelling from left to right with a speed of 80 cm s^{-1} . At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and frequency of the wave.

	amplitude	frequency
A.	6 cm	2 Hz
B.	6 cm	4 Hz
C.	9 cm	2 Hz
D.	9 cm	4 Hz



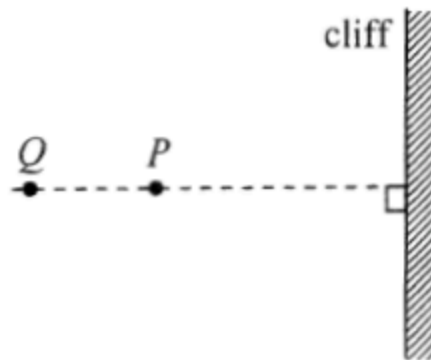
A vibrator generates a stationary wave on a string which is fixed at one end. The figure shows the appearance of the string at a certain instant. Which of the following descriptions about the motion of particles *P*, *Q* and *R* must be correct?

- (1) *P* and *Q* are momentarily at rest at this instant.
- (2) *Q* and *R* take the same time to reach their respective equilibrium positions.
- (3) *P* and *R* are always in antiphase.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

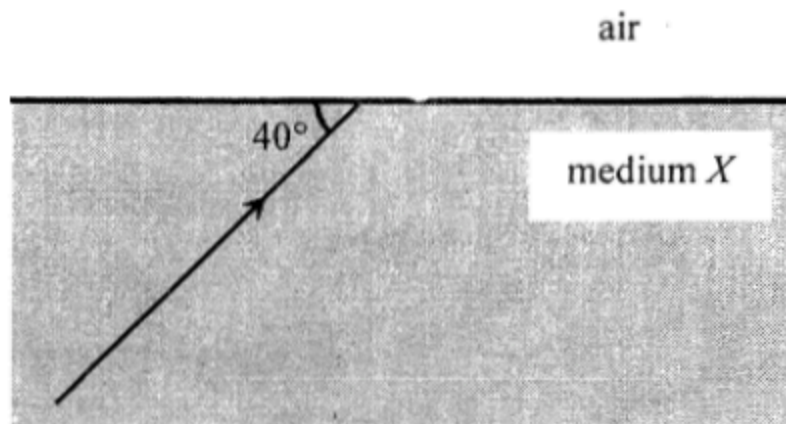
19.

Top view



Astronauts P and Q stand at 400 m and 600 m respectively from a vertical cliff on the surface of a planet. The figure shows the top view. P claps his hands once and Q hears two clapping sounds separated by 4 s. What is the speed of sound in the atmosphere of this planet ?

- A. 100 m s^{-1}
- B. 150 m s^{-1}
- C. 200 m s^{-1}
- D. 250 m s^{-1}



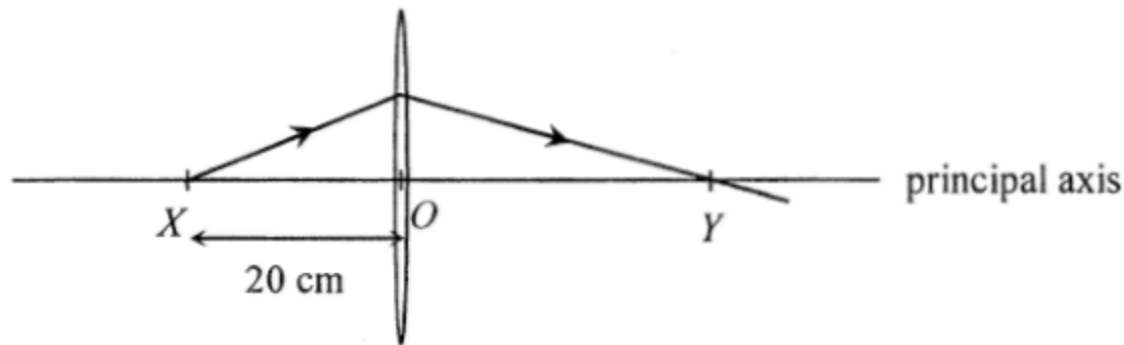
A ray of light is travelling from a transparent medium X to air making an angle of 40° with the boundary plane as shown. If the angle between the refracted ray in air and the reflected ray in medium X is 70° , find the refractive index of medium X .

- A. $\frac{\sin 40^\circ}{\sin 30^\circ}$
- B. $\frac{\sin 30^\circ}{\sin 40^\circ}$
- C. $\frac{\sin 60^\circ}{\sin 50^\circ}$
- D. $\frac{\sin 50^\circ}{\sin 60^\circ}$

21. White light can be resolved into component colours by using a glass prism. Which of the following statements is/are correct ?

- (1) The refractive indices of glass for different component colours are not the same.
- (2) Red light travels faster than violet light in a vacuum.
- (3) The frequencies of all the component colours are reduced when entering the prism.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

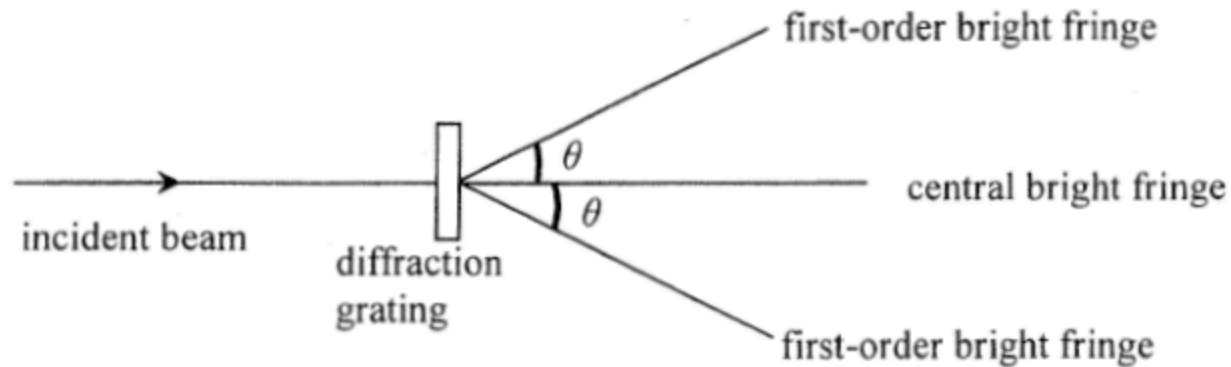


A point light source at X on the principal axis of a thin convex lens emits a ray of light. The ray passes through the lens and reaches the principal axis at point Y as shown. O is the optical centre of the lens such that $OX = 20\text{ cm}$ and $OY > OX$. Which of the following statements is/are correct ?

- (1) The focal length of the lens is shorter than 20 cm .
- (2) If the point light source is shifted away from the lens, separation OY would increase.
- (3) An object placed at Y would give a diminished image at X .

- A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

*23.



When monochromatic light passes through a diffraction grating, a pattern of bright fringes is formed. Which arrangement would produce the greatest angle θ between the central and first-order bright fringes ?

	grating (lines per mm)	colour of light
A.	400	green
B.	400	blue
C.	200	green
D.	200	blue

24. X and Y are two small identical metal spheres carrying charges $-2Q$ and $+6Q$ respectively. When X and Y are separated by a certain distance, the magnitude of the electrostatic force between them is F .

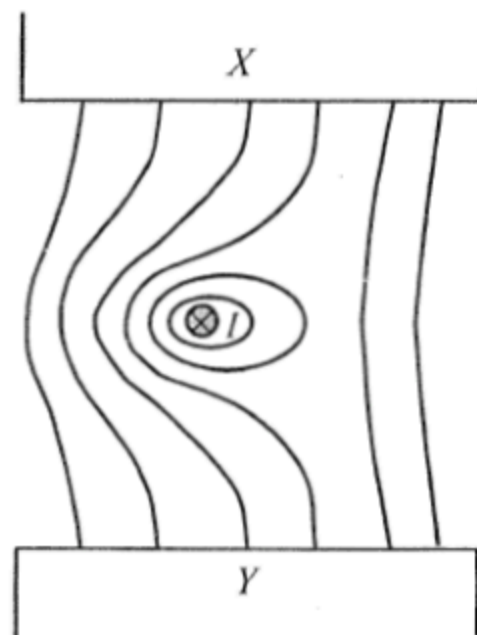


The spheres are brought to touch each other and then placed back to their original positions. The electrostatic force between them becomes

- A. $\frac{1}{4}F$, attractive.
- B. $\frac{1}{4}F$, repulsive.
- C. $\frac{1}{3}F$, attractive.
- D. $\frac{1}{3}F$, repulsive.

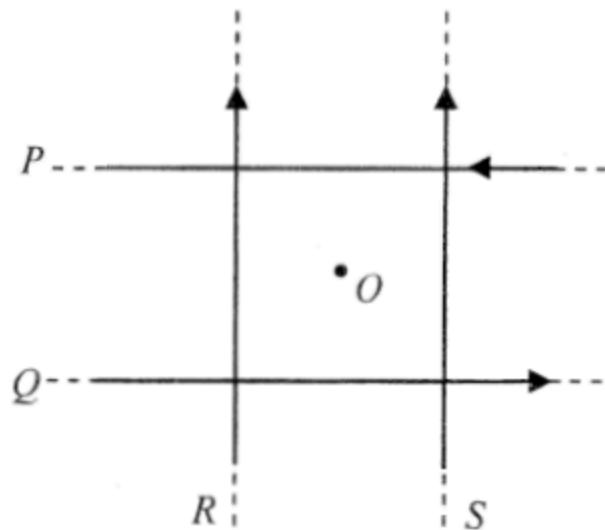
- *25. Lightning flash may occur when the strength of the electric field (assumed uniform) between a thundercloud and the ground reaches $3 \times 10^6 \text{ N C}^{-1}$. A lightning flash on average discharges about 20 C of charge. If a thundercloud is at a height of 500 m above the ground, estimate the order of magnitude of the energy released in a lightning flash.

- A. 10^6 J
- B. 10^8 J
- C. 10^{10} J
- D. 10^{12} J



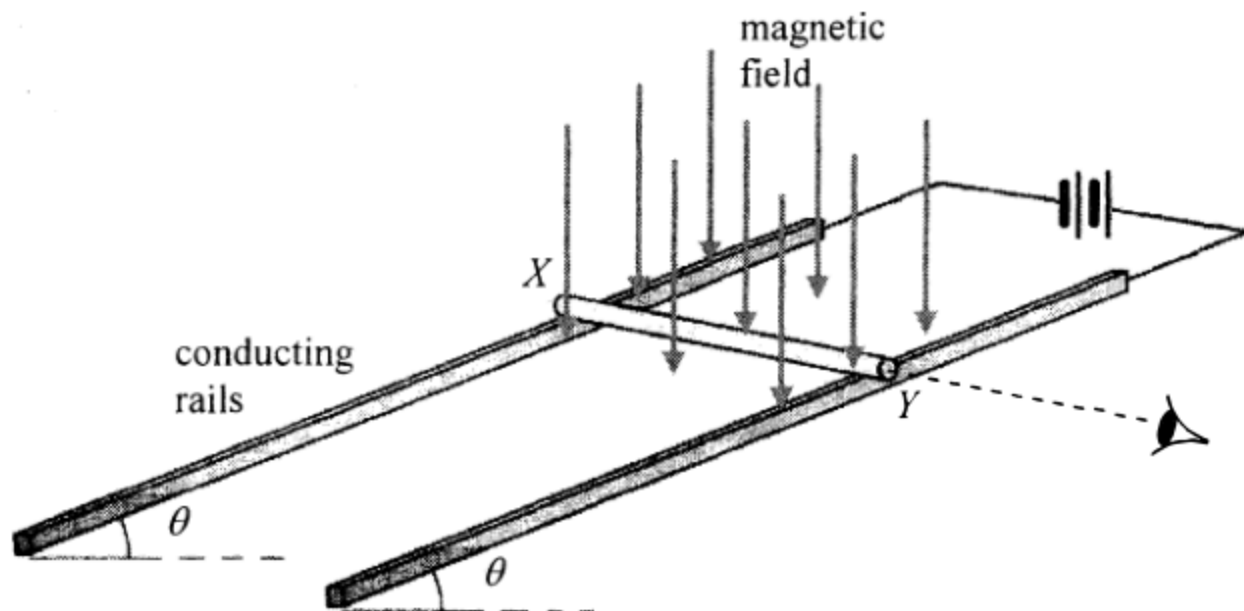
A straight wire carrying current I pointing into the paper is placed in a magnetic field between pole pieces X and Y . The figure shows the resultant field line pattern. What is the polarity of pole piece X and in what direction is the magnetic force acting on the wire? Ignore the effect of the Earth's magnetic field.

	polarity of X	direction of magnetic force
A.	N	to right
B.	N	to left
C.	S	to right
D.	S	to left



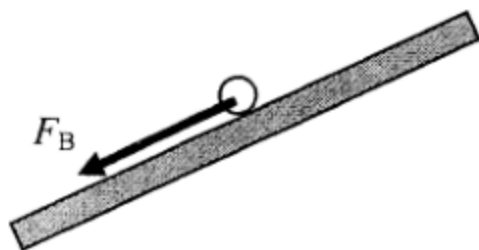
In the figure, four long straight wires P , Q , R and S in the same plane carry equal currents in the directions shown. The wires are insulated from each other. O is a point on the same plane and is equidistant from each wire. Removing which wire would increase the magnetic field strength at O ?

- A. wire P
- B. wire Q
- C. wire R
- D. wire S

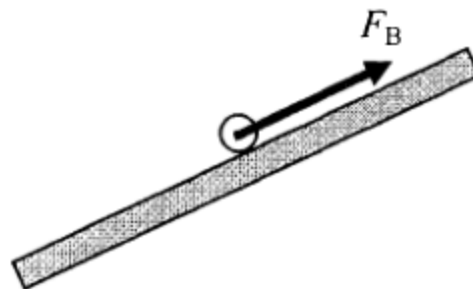


A copper rod XY is placed on a pair of smooth inclined conducting rails which are located in a magnetic field applied vertically downward. The rails make an angle θ to the horizontal and a battery is connected to the rails as shown above. Which diagram shown below represents the magnetic force F_B acting on the rod when viewed from end Y ?

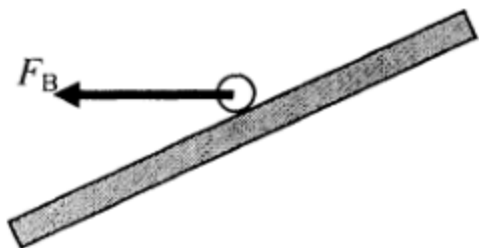
A.



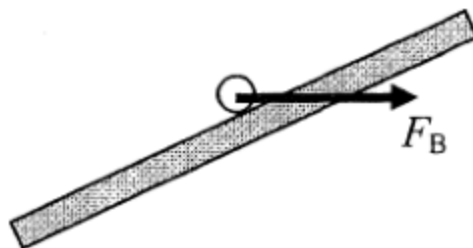
B.



C.



D.



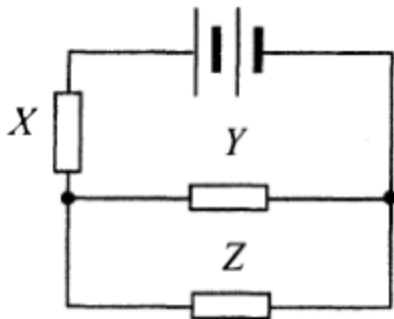
29.



A metal rod OP is rotated about O in a clockwise direction in the plane of the paper with a uniform magnetic field pointing into the paper. Which statement is correct ?

- A. An induced current flows in the rod from O to P .
- B. An induced current flows in the rod from P to O .
- C. E.m.f. is induced in the rod with end O at a higher electric potential.
- D. E.m.f. is induced in the rod with end P at a higher electric potential.

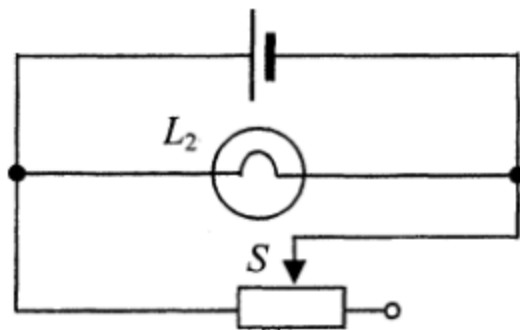
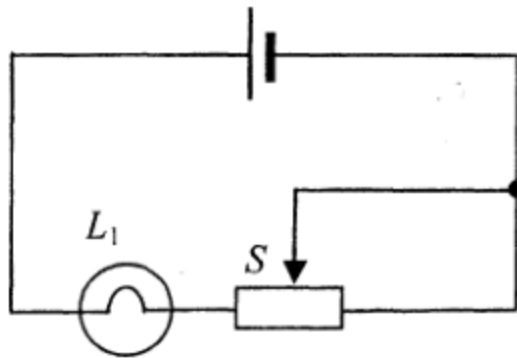
30.



Resistors X , Y and Z in the above circuit are identical while the battery of negligible internal resistance supplies a total power of 24 W. What is the power dissipated in resistor Z ?

- A. 3 W
- B. 4 W
- C. 6 W
- D. 8 W

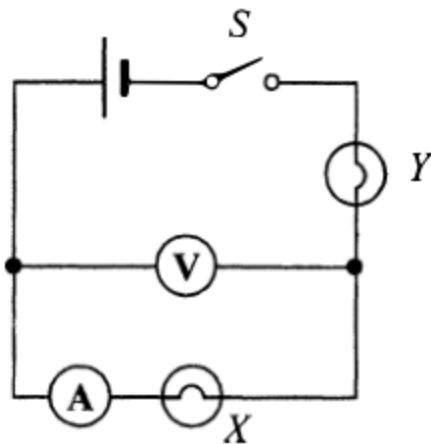
31.



In each of the above circuits, the cell has constant e.m.f. and negligible internal resistance. When the sliding contact S of each rheostat shifts from the mid-position to the right, how would the brightness of each bulb change ?

- | | bulb L_1 | bulb L_2 |
|----|-------------------|-------------------|
| A. | becomes dimmer | remains unchanged |
| B. | becomes dimmer | becomes brighter |
| C. | remains unchanged | becomes dimmer |
| D. | becomes brighter | remains unchanged |

32.



In the above circuit, the cell has negligible internal resistance. When switch S is closed, both bulbs are not lit. The voltmeter has a reading but the ammeter reads zero. If only one fault has been developed in the circuit, which of the following is possible ?

- A. Bulb X has been shorted accidentally.
- B. Bulb Y has been shorted accidentally.
- C. Bulb X is burnt out and becomes open circuit.
- D. Bulb Y is burnt out and becomes open circuit.

33. Which of the following domestic electrical appliances consumes a power close to 1 kW in normal working conditions ?

- A. an electric fan
- B. a microwave oven
- C. a fluorescent lamp
- D. a TV set

34. $^{238}_{92}\text{U}$ undergoes α - β - β - α decay and becomes a nuclide X . What are the atomic number and mass number of X ?

	atomic number	mass number
A.	90	230
B.	90	234
C.	88	230
D.	88	234

*35. Polonium-210 is a pure α -emitter with a half-life of 140 days and it will decay into lead, which is stable. Initially there is a sample containing 420 mg of pure polonium-210. Estimate the mass of polonium-210 left after 70 days.

- A. 315 mg
- B. 297 mg
- C. 210 mg
- D. 105 mg

*36. The sun releases huge amount of energy through thermonuclear fusion while at the same time its mass decreases. The average power released by the sun is about 3.8×10^{26} W. Estimate the decrease in mass of the sun in one second.

- A. 4.2×10^6 kg
- B. 4.2×10^9 kg
- C. 1.3×10^{15} kg
- D. 1.3×10^{18} kg

END OF SECTION A